Entropy Has No Direction: Constraint Engineering Reshapes Entropy Distributions and Paves the Way for Energy Freedom

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The LaTeX source of this manuscript, together with a live FAQ, is available at https://github.com/tpeng1977/entropy.

Abstract

We revisit textbook claims that entropy must increase and show that, under time-reversal invariant microscopic dynamics, no universal trajectory-wise or statistical assertion that the coarse-grained entropy $S(t)$ is non-decreasing can hold. The core is a mirror-state construction: for any microstate A one constructs its time-reversed partner B (momenta inverted); requiring $S(t)$ to be non-decreasing for both A and B forces every time to be a local minimum of S and hence makes $S(t)$ constant along the trajectory. The consistent picture is that entropy is a stochastic variable described by a probability distribution $P(S)$ whose shape depends on constraints and boundary conditions; entropy-based regularities are emergent summaries of constraint-dependent microscopic dynamics, and in practice it is constraints and boundaries—not entropy itself—that one manipulates to achieve mixing, separation, or self-organization. Working with Boltzmann (coarse-grained) entropy on the energy shell, we then derive from first principles how constraints reshape the long-time entropy distribution $P_\infty(S;\lambda)$ by altering the invariant measure through changes in the Hamiltonian and/or the accessible phase space. In the microcanonical setting we obtain a sharp criterion: the *only* way $P_\infty^{(E)}(S;\lambda)$ can remain the same up to translation is when all accessible macrostate volumes are scaled by a common factor; otherwise the distribution changes structurally. We connect this framework to experiments on asymmetric nanopores and molecular gates, to macroscopic examples from civil engineering (windbreak forests, dikes, vortex suppression, traffic-flow control), and to natural phenomena such as lightning guided to lightning rods, snowflake and mineral-veil growth, and the sudden crystallisation of supercooled water. In all these cases, constraints and dynamics together determine the long-time entropy distribution and can make ordered, low-entropy macrostates statistically favoured compared with the unconstrained case.

1 Introduction

To enable safe, comfortable, and rapid travel, we have overcome geological and environmental obstacles, spanned natural chasms, and turned the impossible into the possible—building highways, railways, and airports on a vast scale. These achievements stand as a cornerstone of civilisation. They rest on a single principle: constraints reshape what is accessible, and thus what becomes typical. This paper reconstructs the core cognition of entropy evolution from this perspective of engineering constraint design and shows that the same principle extends from macroscopic engineering to the foundations of thermodynamics. Its most important conclusion is that **constraints change the system's operating characteristics and reshape the entropy distribution**.

The energy crisis has become a core bottleneck restricting the sustainable development of human civilisation. Abundant thermal energy surrounds us; if it can be harnessed and put to fuller use, we may achieve energy freedom; if the universe is destined for heat death, that possibility is foreclosed. Driven by the desire to make fuller use of thermal energy and the refusal to accept the heat-death thesis, we take logic and mathematics as

our tools and embark on this technical exploration. We take as our sole basis four axioms (Sec. 2): Hamiltonian dynamics (Axiom 1), time-reversal invariance (Axiom 2), Boltzmann coarse-grained entropy (Axiom 3), and phase-space and invariant measure (Axiom 4). No other laws or postulates are invoked.

From these four axioms we derive two results and prove that entropy has no intrinsic direction. (i) **Entropy has no direction.** A mirror-state construction (Axioms 1–3) shows that any universal statement of the form "entropy does not decrease forward in time" (trajectory-wise or as a universal statistical principle) is incompatible with the axioms; the derived picture is that entropy is described by a probability distribution $P(S)$, without an intrinsic direction. (ii) **Constraints and boundary conditions reshape this distribution.** From Axiom 4 we prove that constraints reshape the long-time entropy distribution $P_\infty(S;\lambda)$. Constraints, as the core variable of engineering design, thus relax the long-standing limit of entropy increase and pave a brand-new path for human energy freedom; the main takeaway is that constraints change the system's operating characteristics and reshape the entropy distribution. The proof of (ii) was motivated in part by the author's earlier simulation work showing that geometry can alter entropy regimes in nanofluidic cascades [7]. We then discuss recent experimental work that validates this framework. This research challenges traditional thermodynamic fatalism and opens the era of thermodynamic design, which is of profound significance to the long-term survival and development of human beings.

2 Axioms

The entire paper relies only on the following four axioms of classical mechanics and statistical mechanics. No other laws or postulates are used.

Axiom 1 (Hamiltonian dynamics). The microscopic evolution of an isolated system obeys classical Hamiltonian dynamics. The phase-space microstate $\Gamma(t)$ satisfies Hamilton's equations $\dot{\Gamma} = \{H, \Gamma\}$, where H is the system Hamiltonian and $\{\cdot,\cdot\}$ denotes the Poisson bracket.

Axiom 2 (Time-reversal invariance). There exists a time-reversal operator $\mathcal{T}$ (for classical mechanics: reverse all momenta, leave positions unchanged). If $\Gamma_A(t)$ is a solution of Hamilton's equations, then the mirror state $\Gamma_B(t) := \mathcal{T}\Gamma_A(2t_0 - t)$ is also a solution of the same Hamilton's equations for any fixed t_0. In particular, the forward-time evolution of $\Gamma_B(t_0) = \mathcal{T}\Gamma_A(t_0)$ replays the past of Γ_A in reverse.

Axiom 3 (Boltzmann coarse-grained entropy). Phase space is partitioned by a time-reversal symmetric coarse-graining $\{C_m\}_{m\in\mathcal{M}}$ (i.e. $\mathcal{T}C_m = C_m$ for each macrostate C_m). The Boltzmann entropy of macrostate C_m is $S_m = k_B \ln(W_m/W_0)$, where W_m is the phase-space volume of C_m and $W_0 > 0$ is an arbitrary reference volume (shifting by a constant does not affect any conclusions). The instantaneous entropy of the system is $S(\Gamma) = S_m$ if and only if $\Gamma \in C_m$; equivalently, $S(\mathcal{T}\Gamma) = S(\Gamma)$.

For the derivation of $P_\infty(S;\lambda)$ we use the energy-shell form: at fixed energy E, $W_m^{(E)}$ is the accessible volume of C_m on the energy shell, and $S_m^{(E)} = k_B \ln(W_m^{(E)}/W_0)$.

Axiom 4 (Phase space and invariant measure). Phase-space evolution satisfies Liouville's theorem (measure is conserved). The long-time invariant measure is determined solely by the Hamiltonian and the accessible phase space: for an isolated system at fixed energy E, the invariant measure is microcanonical (uniform on the accessible energy shell); for a system coupled to a heat bath at temperature T, the invariant measure is canonical (Gibbs with that Hamiltonian and $\beta = 1/(k_B T)$).

Remark (continuous or discrete). The mirror-state argument below holds *whether* $S(t) := S(\Gamma(t))$ *is continuous along trajectories or not.* If it is continuous (smooth coarse-graining), then "every time is a local minimum" implies $S(t)$ is constant. If it is piecewise constant with jumps (discrete coarse-graining), Eq. (1) still forces $S(t_0)$ to be a two-sided minimum at every t_0, ruling out any jump up or down, so again $S(t)$ is constant. The conclusion is the same in both cases.

2.1 Interpretation of $P_\infty(S;\lambda)$

In the second part of the paper, $P_\infty(S;\lambda)$ denotes the *long-time* entropy distribution induced by the invariant measure of Axiom 4. Under ergodicity or when the initial ensemble is drawn from the invariant measure, P_∞ is the limiting distribution of S over long time. The formulas below are statements about these invariant measures and the entropy distributions they induce.

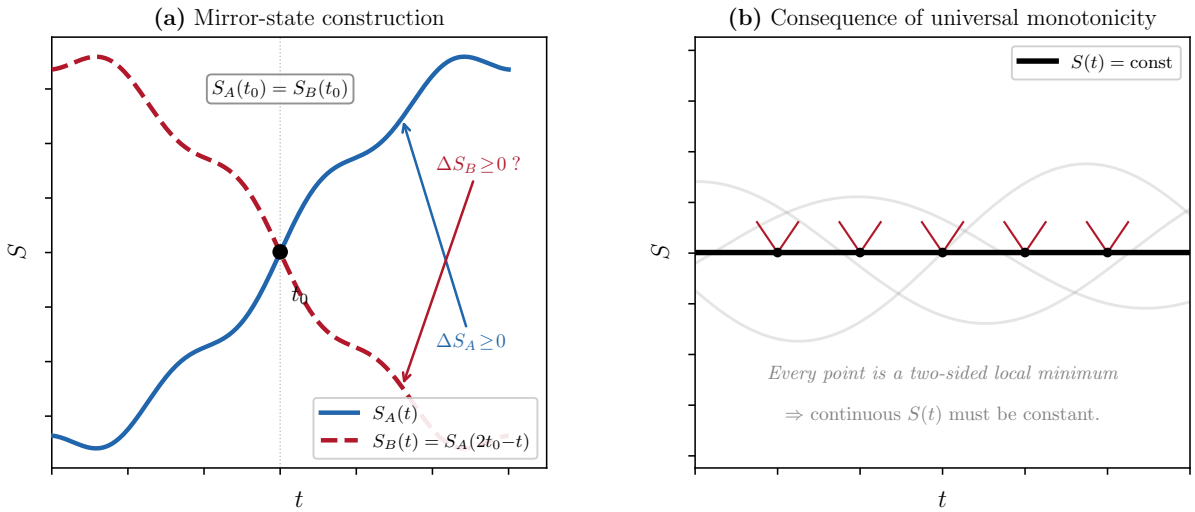

Figure 1: The mirror-state paradox. **(a)** For any trajectory $S_A(t)$ (solid blue) and its time-reversed partner $S_B(t) = S_A(2t_0 - t)$ (dashed red), universal monotonicity would require both $\Delta S_A \geq 0$ and $\Delta S_B \geq 0$ forward from t_0. Because $S_B(t_0+\delta t) = S_A(t_0-\delta t)$, the second condition forces S_A to be non-decreasing *backward* as well, making t_0 a two-sided local minimum. **(b)** Since t_0 is arbitrary, every point must be a local minimum; whether $S(t)$ is continuous or not, this implies $S(t) = \text{const}$, contradicting observed entropy changes.

3 The mirror-state paradox

3.1 Strict (trajectory-wise) monotonicity is impossible as a universal law

Claim. From Axioms 1–3 (Sec. 2), a universal statement of the form "for every microstate and every time, $S(t + \delta t) \geq S(t)$ for all sufficiently small $\delta t > 0$" is incompatible.

Proof (mirror-state construction). Fix an arbitrary time t_0 on an arbitrary trajectory $\Gamma_A(t)$. Construct the mirror state at the same time: $\Gamma_B(t_0) = \mathcal{T}\Gamma_A(t_0)$, and let $\Gamma_B(t)$ be its forward-time evolution. By Axiom 2, for any $\delta t > 0$,

$$\Gamma_B(t_0 + \delta t) = \mathcal{T}\Gamma_A(t_0 - \delta t).$$

By Axiom 3, $S(\mathcal{T}\Gamma) = S(\Gamma)$, so

$$S_B(t_0 + \delta t) = S_A(t_0 - \delta t), \qquad S_B(t_0) = S_A(t_0).$$

Now apply the universal monotonicity statement to *both* A and B:

$$S_A(t_0 + \delta t) \geq S_A(t_0), \qquad S_B(t_0 + \delta t) \geq S_B(t_0).$$

The second inequality becomes $S_A(t_0 - \delta t) \geq S_A(t_0)$. Hence, for all small $\delta t > 0$,

$$S_A(t_0 + \delta t) \geq S_A(t_0) \quad \text{and} \quad S_A(t_0 - \delta t) \geq S_A(t_0). \tag{1}$$

Thus t_0 is a (two-sided) local minimum of $S_A(t)$.

Because t_0 was arbitrary, *every* time is a local minimum. Whether $S_A(t)$ is continuous or piecewise constant (Sec. 2), a function for which every time is a two-sided local minimum must be constant. Therefore any universal statement that entropy does not decrease along every trajectory would imply that entropy is constant on every trajectory, eliminating any entropic arrow of time. □

3.2 Extension: universal statistical monotonicity is likewise incompatible

The same mirror-state logic shows that a *universal* statistical principle "entropy increase is overwhelmingly probable for the forward-time evolution of any initial microstate" is incompatible with time-reversal invariance. Apply that principle to A at time t_0, and also to $B = \mathcal{T}A$ at t_0. Since the forward evolution of B corresponds to the time-reversed past of A, "overwhelmingly probable increase" for both implies (with overwhelming probability) the two-sided inequalities in Eq. (1), i.e. that t_0 is a local minimum with overwhelming probability. Choosing t_0 arbitrarily removes any persistent arrow of time in the same way.

Thus, from Axioms 1–3, no universal one-way statement about ΔS can hold for every microstate; any such statement must depend on additional structure (initial ensemble, coarse-graining, limits, or constraints/boundaries). This leads to the derived view below.

4 Derived view: entropy is a random variable with a constraint-dependent PDF

From the preceding derivation, the picture is: **entropy has no direction; it has a probability distribution**. Write $P_t(S)$ for the (time-dependent) entropy distribution induced by an ensemble of microstates at time t, and $P_\infty(S;\lambda)$ for the long-time distribution under constraint parameters λ (geometry, boundary conditions, fields, etc.). Here and throughout, λ is understood in a broad sense: all forces, interactions, rules, couplings, coordination patterns, and driving protocols that shape the system's dynamics are treated as constraints in this sense. In particular, constraints (including driving) may be static or dynamic (time-dependent); the definitions and conclusions apply whenever the stated axioms/assumptions hold for the corresponding constrained dynamics and the invariant (or long-time) measures they induce.

At the microscopic level, however, nothing in the dynamics "sees" entropy. Individual particles obey time-reversal invariant equations of motion; their trajectories and interactions are governed by the Hamiltonian and forces, not by macroscopic state functions such as entropy. Boltzmann entropy is a coarse-grained, statistical description of the collective behaviour of many particles—a macroscopic *appearance* of the underlying time-reversal invariant motion, obtained only after we partition phase space into macrostates and count accessible microstates. Constraints cannot change the fundamental microscopic rules; their core role is to intervene directly in the actual motion of particles by changing the Hamiltonian $H(\Gamma;\lambda)$ and/or the accessible set $\mathcal{A}(\lambda)$, thereby restricting where particles can go and how they move. This, in turn, changes the number of accessible microstates for each macrostate and thus reshapes the entropy distribution. All changes in entropy are therefore *statistical summaries* of how constraints redirect microscopic trajectories. Entropy is a diagnostic, not a causal driver. In practical engineering terms, what one truly controls and optimises are constraints and boundary conditions (to achieve, say, separation or mixing), while entropy is useful as a bookkeeping tool for whether a given design makes certain macrostates typical or rare. From a logical perspective it is more meaningful to study how different constraints alter system dynamics and long-time distributions than to postulate universal laws about "entropy increase" detached from the underlying microphysics.

From this angle, familiar macroscopic "entropy increase" phenomena are best understood as follows. In many textbook and laboratory situations, the initial state is prepared to be *very special* and low-entropy (gas in one half of a box, sharp temperature gradients, unmixed components, etc.). Starting from such atypical initial conditions, the entropy—as a coarse-grained statistic of particle configurations—does indeed rise rapidly toward the values typical of the long-time distribution under the given constraints. Once the system has reached that regime, however, the entropy does not freeze at a single "equilibrium value": microscopic dynamics continue indefinitely and $S(t)$ keeps fluctuating, with spontaneous increases and decreases both being normal. The role of constraints is to determine the *shape* of the long-time distribution for S (and for other observables), not to enforce monotone drift toward a uniquely defined maximum. There need not be a strict equilibrium state in the sense of a single entropy value, nor must the formal "maximum entropy" compatible with constraints coincide with the most probable coarse-grained macrostate once dynamical restrictions are taken seriously.

What fundamentally decides whether a system mixes, separates, or self-organises into ordered patterns is therefore the constraint-dependent microscopic dynamics and the resulting invariant measures. All results in this paper concerning entropy distributions and their evolution should thus be read as probabilistic, ensemble-level characterisations of typical behaviour under specified constraints, rather than as per-trajectory guarantees for every single realisation of a physical system.

The remainder of this paper proves the second core claim—**constraints and boundaries can change the long-time entropy distribution** $P_\infty(S;\lambda)$ in an explicit, first-principles way—and then presents experimental validation of this framework.

5 First-principles derivation: constraints $\lambda \to P_\infty(S;\lambda)$

This section uses Axioms 1, 3 and 4 to derive how constraint parameters λ determine the long-time entropy distribution $P_\infty(S;\lambda)$.

5.1 Constraints as Hamiltonian/accessible-set modifications

Let λ denote constraint parameters. Constraints enter through the Hamiltonian

$$H(\Gamma;\lambda) = H_0(\Gamma) + V_c(\Gamma;\lambda), \tag{2}$$

and/or through an accessible set $\mathcal{A}(\lambda)$ (hard constraints). Both views are equivalent: hard walls correspond to $V_c = +\infty$ outside $\mathcal{A}(\lambda)$.

5.2 Long-time invariant measures (microcanonical and canonical)

By Axiom 4, the invariant measure is determined solely by the Hamiltonian and the accessible phase space. We make this explicit for the two standard cases.

Microcanonical (isolated). At fixed energy E, the invariant measure is uniform on the accessible energy shell:

$$\rho_\infty^{(E)}(\Gamma;\lambda) = \frac{1}{\Omega(E;\lambda)}\,\delta\big(H(\Gamma;\lambda)-E\big)\,\mathbf{1}_{\mathcal{A}(\lambda)}(\Gamma), \tag{3}$$

where the accessible density of states is

$$\Omega(E;\lambda) = \int \mathrm{d}\Gamma\,\delta\big(H(\Gamma;\lambda)-E\big)\,\mathbf{1}_{\mathcal{A}(\lambda)}(\Gamma). \tag{4}$$

Canonical (heat bath). With a heat bath at temperature T and dynamics obeying fluctuation–dissipation, the invariant measure is

$$\rho_\infty(\Gamma;\lambda) = \frac{1}{Z(\lambda)}\,e^{-\beta H(\Gamma;\lambda)}, \qquad Z(\lambda) = \int_{\Gamma\in\mathcal{A}(\lambda)} \mathrm{d}\Gamma\,e^{-\beta H(\Gamma;\lambda)}. \tag{5}$$

5.3 Boltzmann entropy on the energy shell

Fix a time-reversal symmetric coarse-graining $\{C_m\}_{m\in\mathcal{M}}$ (Axiom 3). Define the accessible macrostate volume on the energy shell:

$$W_m^{(E)}(\lambda) = \int \mathrm{d}\Gamma\,\delta\big(H(\Gamma;\lambda)-E\big)\,\mathbf{1}_{C_m}(\Gamma)\,\mathbf{1}_{\mathcal{A}(\lambda)}(\Gamma), \tag{6}$$

and assign the Boltzmann entropy value

$$S_m^{(E)}(\lambda) = k_B \ln\left(\frac{W_m^{(E)}(\lambda)}{W_0}\right). \tag{7}$$

5.4 Closed-form expression for $P_\infty(S;\lambda)$ and a sharp change criterion

Microcanonical. Because the microcanonical measure is uniform on the energy shell, the macrostate probability is a ratio of accessible volumes:

$$\pi_m^{(E)}(\lambda) = \frac{W_m^{(E)}(\lambda)}{\Omega(E;\lambda)}, \qquad \Omega(E;\lambda) = \sum_{m\in\mathcal{M}} W_m^{(E)}(\lambda). \tag{8}$$

Therefore the long-time entropy distribution at fixed energy is

$$\boxed{P_\infty^{(E)}(S;\lambda) = \sum_{m\in\mathcal{M}} \frac{W_m^{(E)}(\lambda)}{\Omega(E;\lambda)}\,\delta\big(S - S_m^{(E)}(\lambda)\big).} \tag{9}$$

Proposition (sharp criterion; "only translation" degeneracy). Fix E and the coarse-graining. Let $\mathcal{V}(\lambda)$ denote the multiset of accessible macrostate volumes $\{W_m^{(E)}(\lambda)\}_{m\in\mathcal{M}}$ (counting multiplicity).

1. If there exists $c>0$ such that $\mathcal{V}(\lambda_2) = c\,\mathcal{V}(\lambda_1)$ (i.e. all macrostate volumes scale by a common factor, up to permutation), then
$$P_\infty^{(E)}(S;\lambda_2) = P_\infty^{(E)}(S - k_B\ln c\,;\lambda_1)\,,$$
i.e. the distribution changes only by translation of the entropy axis.
2. Otherwise, $P_\infty^{(E)}(S;\lambda_2)$ is *not* a translate of $P_\infty^{(E)}(S;\lambda_1)$; the distribution changes structurally.

Proof. Let M be the macrostate index with $\mathbb{P}(M=m) = \pi_m^{(E)}(\lambda)$ and define $X_\lambda := W_M^{(E)}(\lambda)$. Then $S = k_B\ln(X_\lambda/W_0)$ and Eq. (9) is exactly the law of S. If $W_m^{(E)}(\lambda_2) = c\,W_{\sigma(m)}^{(E)}(\lambda_1)$ for some permutation σ, then $X_{\lambda_2} \overset{d}{=} c\,X_{\lambda_1}$, hence $S_{\lambda_2} \overset{d}{=} S_{\lambda_1} + k_B\ln c$, which is the claimed translation. Conversely, if $P_\infty^{(E)}(S;\lambda_2)$ were a translate of $P_\infty^{(E)}(S;\lambda_1)$ by Δ, then $X_{\lambda_2} = W_0 e^{S/k_B}$ would be distributed as $e^{\Delta/k_B}X_{\lambda_1}$, which implies $\mathcal{V}(\lambda_2) = e^{\Delta/k_B}\mathcal{V}(\lambda_1)$ (up to permutation). □

Canonical as an exact energy mixture. The canonical energy density is

$$P_\beta(E;\lambda) = \frac{\Omega(E;\lambda)\,e^{-\beta E}}{Z(\lambda)}, \qquad Z(\lambda) = \int \mathrm{d}E\,\Omega(E;\lambda)\,e^{-\beta E}. \tag{10}$$

Since $P_\infty^{(E)}(S;\lambda)$ is the entropy distribution conditioned on energy E, the canonical long-time entropy distribution is the exact mixture

$$\boxed{P_{\infty,\beta}(S;\lambda) = \int \mathrm{d}E\;P_\beta(E;\lambda)\,P_\infty^{(E)}(S;\lambda).} \tag{11}$$

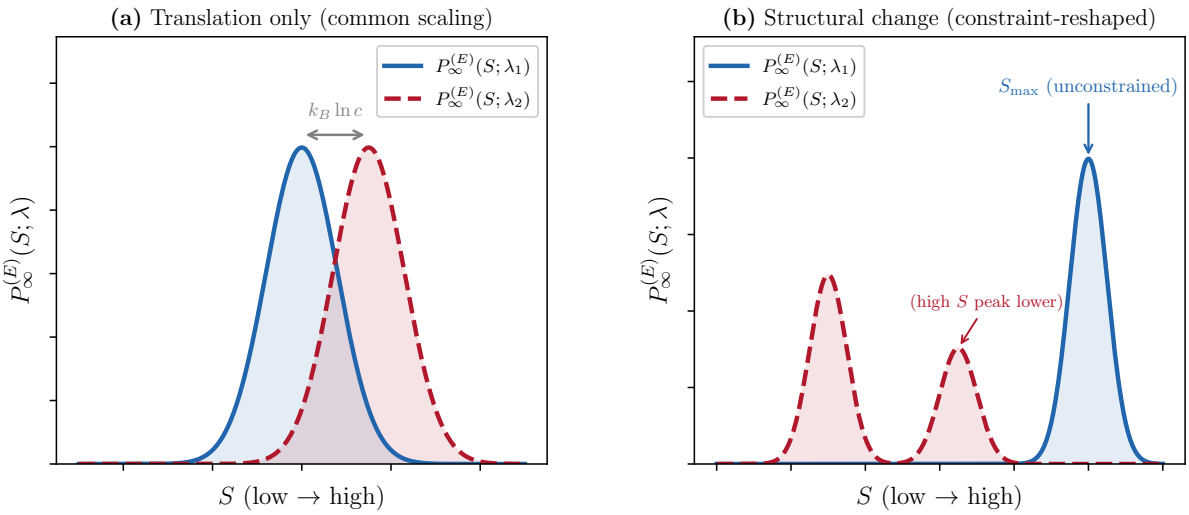

Figure 2: How constraints reshape the long-time entropy distribution $P_\infty^{(E)}(S;\lambda)$. Here P_∞ is the *distribution* of entropy over long time; the system keeps fluctuating and there is no single equilibrium state. **(a)** When all macrostate volumes scale by a common factor c, the distribution translates along the S-axis by $k_B \ln c$ without changing shape (the only degenerate case in the sharp criterion). **(b)** Unconstrained case: entropy is almost entirely concentrated near the *maximum* $S_{\max}$ (narrow peak at high S). Under a generic constraint the distribution reshapes: significant probability mass moves to lower S, making lower-entropy macrostates statistically accessible.

5.5 Bulletproof qualifier: changing rates is not changing the invariant measure

The above results concern changes that alter the invariant measure by changing $\mathcal{A}(\lambda)$ and/or $H(\Gamma;\lambda)$. By contrast, a purely kinetic modification that only rescales crossing rates or mixing times *while leaving the invariant measure unchanged* can change the finite-time distribution $P_t(S)$ but does not change the long-time distribution $P_\infty(S)$.

6 Experimental validation: asymmetric constraints enable spontaneous low-entropy transitions

Recent experimental work by Qiao and Wang [6] provides compelling validation of the theoretical framework presented above. Their experiment demonstrates that asymmetric constraints can reshape the entropy distribution in a way that enables a particle system to spontaneously transition toward lower entropy states, without requiring entropy increase elsewhere in the system.

6.1 The Qiao–Wang experiment

Qiao and Wang investigated nanoporous carbon electrodes in dilute aqueous CsPiv solutions where the effective nanopore size ($d_e \approx 1\,\mathrm{nm}$) only slightly exceeds the ion size ($d_i \approx 0.7\,\mathrm{nm}$), satisfying $d_i < d_e < 2d_i$. The steady-state ion distribution is intrinsically non-equilibrium: the measured $|\delta V|$ is nearly an order of magnitude above the upper limit that equilibrium thermodynamics would assign to such a system, and the system produces useful work in an isothermal cycle by absorbing heat from a single thermal reservoir.

6.2 Interpretation: constraint-reshaped $P_\infty(S;\lambda)$

In our framework the nanopore walls act as the constraint parameter λ that reshapes $\mathcal{A}(\lambda)$. The quasi-one-dimensional confinement fundamentally alters $W_m^{(E)}(\lambda)$: ion trajectories lose full chaoticity, collisions become sparse, and the system cannot relax to global equilibrium—a structural change in $P_\infty^{(E)}(S;\lambda)$ per the sharp criterion of Sec. 5. The observed non-Boltzmannian surface-ion density $\sigma^\pm$ is a direct manifestation: the constraint prevents the system from reaching S_{eq} and instead pins it at $S_{\mathrm{ne}} < S_{\mathrm{eq}}$, without requiring compensating entropy increase elsewhere. It is the entropy landscape itself that has been reshaped, not the entropy balance between subsystems.

6.3 Further experimental support from the literature

Independent work by other groups is consistent with the same mechanism. Ramirez, Gomez, Mafe and coworkers [8, 9] showed that asymmetric conical nanopores rectify a zero-mean fluctuating voltage into a net ionic current, charging an external capacitor to about 1 V; the constraint λ (pore geometry) makes the long-time charge-transfer distribution asymmetric under sign flip of the drive. Qiao, Shang, and Kou [10] demonstrated a molecular-sized gate (locally nonchaotic barrier) in an isolated system, where gas spontaneously flows from the low-pressure to the high-pressure side and useful work can be extracted in a cycle from a single thermal reservoir—the same constraint-reshaping picture in a different geometry. Powell, Vlassiouk, and Siwy [11] reported voltage-polarity-dependent nonequilibrium 1/f noise in rectifying nanopores (Phys. Rev. Lett.): under one polarity the system exhibits equilibrium-like fluctuations, under the other distinctly non-equilibrium fluctuations, illustrating how λ shapes not only the mean current but the fluctuation spectrum. Together with the Qiao–Wang result, these experiments show that constraint-reshaped entropy distributions are not an isolated finding but a recurring pattern across nanoscale systems.

In the same spirit, crystal semiconductor diodes provide a simple asymmetric-constraint platform that can satisfy the prerequisite for directional rectification of thermal motion at the circuit level. Our preliminary tests in non-specialized laboratory conditions indicate a stable polarity of the generated potential difference; when diodes are connected in series with the same orientation, the potential difference increases, while opposite orientation leads to mutual cancellation.

Technical outlook: unidirectional nanopore membranes and phase change. In the present framework, **unidirectional (asymmetric) nanopore membranes** and **phase change** are two approaches for which the hope of achieving **sustained useful work or power generation from a single heat source** is relatively high. Asymmetric pores already rectify flows and reshape $P_\infty(S;\lambda)$ so that net work from one reservoir is possible; phase change (e.g. liquid–vapour or subcooled liquid flash) provides large density and entropy changes that can be harnessed under the same constraint-reshaping mechanism. Together they are among the schemes with relatively high hope: nanopore-like asymmetry (or equivalent constraints) plus a phase-change working fluid can in principle yield continuous, single-reservoir power output, consistent with the theory developed in this paper and with the companion implementation (micro temperature difference, subcooled liquid, liquid-only compression) outlined elsewhere.

6.4 Macroscopic engineering validation: Constraint-induced entropy shaping in civil engineering and transport

Beyond the microscopic nanopore experiment, the framework of constraint-reshaped $P_\infty(S;\lambda)$ has been implicitly validated by the long history of practice in civil engineering and transport (the author's core research field). For desert roads, asymmetric windbreak forest belts (constraint parameter λ_1) act as geometric boundaries that reshape the phase space of sand particle motion: the accessible macrostate volume for sand deposition on the pavement is drastically reduced, while the volume for sand accumulation outside the forest belt is expanded. This structural change in macrostate volumes (per the sharp criterion in Sec. 5) reshapes the long-time entropy distribution of the sand–wind system, making the low-entropy state (road unobstructed, ordered sand motion) the statistically dominant macrostate instead of the high-entropy random sand deposition.

Similarly, for river-crossing roads, spur dikes and longitudinal dikes (constraint parameter λ_2) alter the Hamiltonian of water flow by changing the flow field and energy dissipation, reshaping the entropy distribution of the water–sediment system. The constrained flow field reduces the accessible macrostate volume for high-energy scouring of bridge piers, shifting the long-time distribution $P_\infty(S;\lambda_2)$ toward lower entropy (stable flow, no structural damage). In both cases, these civil engineering and transport measures do not "fight against entropy increase" in the traditional thermodynamic sense, but instead design specific constraints to reshape the entropy landscape—the same core mechanism as the nanopore experiment, just scaled to macroscopic geophysical systems.

These engineering practices confirm that constraint-reshaped entropy distribution is not a microscopic peculiarity but a universal principle applicable to all complex systems, from nanoscale ion motion to macroscale geophysical dynamics. Windbreaks, dikes, vortex suppression, and vehicle–road coordination have repeatedly achieved ordered, low-entropy outcomes through constraint design alone.

Natural phenomena provide further examples of the same mechanism. Lightning strikes are guided with high spatial precision toward lightning rods because the conducting structure reshapes the electric field and the accessible paths for charge motion. Snowflake growth, the formation of mineral veins, and the sudden crystallisation of supercooled water all involve geometric and chemical constraints that guide microscopic dynamics from relatively disordered configurations into highly ordered crystalline structures. In each case, constraints determine the long-time entropy distribution and make these low-entropy macrostates statistically favoured compared with the unconstrained case.

7 Conclusion

The most important conclusion of this work is that constraints change the system's operating characteristics and reshape the entropy distribution. Constraints fix the accessible phase space and the Hamiltonian (and thus how the system evolves); they thereby alter the invariant measure and the long-time entropy distribution $P_\infty(S;\lambda)$. This is proved from the four axioms of Sec. 2, with a sharp microcanonical criterion that isolates the only degenerate case of mere translation. Entropy has no intrinsic direction: it is described by a probability distribution $P(S)$ that depends on constraints and boundary conditions, as shown by the mirror-state construction (Axioms 1–3). This work was driven by the desire for energy freedom and the refusal to accept the heat-death thesis.

A concise way to summarise the conceptual stance of this work is: *constraints set the landscape* (they fix the accessible phase space and the Hamiltonian, and thus which macrostates are possible); *dynamics set the rules* (time-reversal invariant microscopic evolution governs how particles move, independent of entropy); *stochasticity—or in quantum mechanics, intrinsic indeterminacy—determines which of the accessible states is realised*; and *entropy is only a reading*, a coarse-grained statistic of the current macrostate, not a driver of the dynamics. In that picture, the future is not predetermined by a single entropic fate: under different constraints, different futures—mixing, separation, or self-organization—become possible, and each regime remains open to design. Every day is new.

This pattern—constraints reshaping the entropy landscape and enabling spontaneous formation of structure—is familiar from star and planet formation, crystal and snowflake growth, mineral veins, and arguably the emergence of life: all instances of constraint-induced entropy decrease, consistent with the present framework. It is also the core principle behind human engineering design in civil engineering and transport. Windbreak forests for desert roads, spur dikes for river-crossing roads, aerodynamic constraints for bridge vortex vibration suppression, and vehicle–road coordination technology for smoother traffic flow all achieve ordered, low-entropy system states (safe, unobstructed, and fluid traffic) by reshaping the entropy distribution of geophysical and traffic systems (sand–wind, water–sediment, wind–bridge coupling, and vehicle–road flow).

In this framework, whether a system can produce useful work in a cycle by absorbing heat from a single thermal reservoir is determined by the constraints and the resulting $P_\infty(S;\lambda)$; it is a limit-dependent, model-dependent question. The possibility of harnessing abundant thermal energy through constraint design is in that sense open in principle.

The experimental validation by Qiao and Wang [6] demonstrates that this possibility is not merely theoretical but has been realised. Their work shows that asymmetric constraints can reshape the entropy distribution to enable spontaneous low-entropy transitions, allowing a system to produce useful work in a cycle by absorbing heat from a single thermal reservoir. With this experimental confirmation, the fuller exploitation of thermal energy is no longer only consistent with the derived framework but has been achieved in the laboratory.

The path forward is clear: through iterative enhancement, engineering optimisation, and scaling of constraint-based entropy reshaping, practical devices can be developed. In particular, **unidirectional nanopore membranes** (or analogous asymmetric constraints) and **phase change** are two approaches for which the hope of realising sustained useful work or power generation from a single heat source is relatively high, as argued in Sec. 6. The implications extend far beyond energy generation. Because constraints can perpetually reshape entropy distributions and enable spontaneous order, the macroscopic world need not tend toward a featureless equilibrium: universal heat death is not a foregone conclusion but an artefact of the unconstrained limit. With this principle at hand, humanity can sustain civilisation even through extended ice ages, ensuring the perpetual flourishing and development of human culture. The macroscopic world will not end in silence; it will continuously manifest various wonderful phenomena, sustained by the ever-present possibility of constraint-induced order generation.

The era of thermodynamic fatalism is giving way to the age of thermodynamic design. The end of thermodynamic fatalism means that human beings are no longer passive acceptors of thermodynamic laws, but active designers of entropy evolution. Through constraint engineering, we can not only realise energy freedom, but also shape the ordered evolution of various complex systems, from nanoscale energy devices to macroscale geophysical systems, and even the entire universe. This research points to a shift in which engineering science drives the development of fundamental physics and human civilisation extends the scope of what is possible within natural laws. That outcome answers the desire for energy freedom and the refusal to accept the heat-death thesis with which we began.

The arguments and proofs in this paper are formulated entirely within the framework of classical physics (classical mechanics and classical statistical mechanics). In other areas of physics, whenever analogous assumptions about microscopic reversibility, coarse-grained entropy, and constraint-induced reshaping of the invariant measure are satisfied, one may expect results that are equivalent or closely analogous in spirit.

References

[1] H. Goldstein, C. Poole, and J. Safko, *Classical Mechanics* (Addison–Wesley, 3rd ed., 2002).

[2] H. B. Callen, *Thermodynamics and an Introduction to Thermostatistics* (Wiley, 2nd ed., 1985).

[3] R. K. Pathria and P. D. Beale, *Statistical Mechanics* (Elsevier, 3rd ed., 2011).

[4] R. Kubo, M. Toda, and N. Hashitsume, *Statistical Physics II: Nonequilibrium Statistical Mechanics* (Springer, 2nd ed., 1991).

[5] J. L. Lebowitz, "Boltzmann's entropy and time's arrow," *Physics Today* **46**(9), 32–38 (1993).

[6] Y. Qiao and M. Wang, "Breaking the boundaries of the second law of thermodynamics: Intrinsic nonequilibrium distribution of large ions in charged small nanopores," arXiv:2407.04599 [cond-mat.soft] (2025).

[7] T. Peng, "Geometry challenges entropy: Regime-dependent rectification in nanofluidic cascades," arXiv:2602.13931 [physics.comp-ph] (2026).

[8] P. Ramirez, V. Gomez, J. Cervera, S. Nasir, M. Ali, W. Ensinger, and S. Mafe, "Energy conversion from external fluctuating signals based on asymmetric nanopores," *Nano Energy* **16**, 375–382 (2015).

[9] V. Gomez, P. Ramirez, J. Cervera, S. Nasir, M. Ali, W. Ensinger, and S. Mafe, "Charging a capacitor from an external fluctuating potential using a single conical nanopore," *Sci. Rep.* **5**, 9501 (2015).

[10] Y. Qiao, Z. Shang, and R. Kou, "Molecular-sized outward-swinging gate: Experiment and theoretical analysis of a locally nonchaotic barrier," *Phys. Rev. E* **104**, 064133 (2021); arXiv:2106.06648.

[11] M. R. Powell, I. Vlassiouk, C. Martens, and Z. S. Siwy, "Nonequilibrium 1/f noise in rectifying nanopores," *Phys. Rev. Lett.* **103**, 248104 (2009).